

Probability & Random Processes

EE-2011 (BEE-4B)

Department of Electrical Engineering

2.4 & 2.5 **Probability of an Event** **Additive Rules**

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2.4 Probability of an event

- Throughout the remainder of this chapter, we consider only those experiments for which the sample space contains a finite number of elements.
- The *likelihood of the occurrence of an event*
- Resulting from such a statistical experiment is evaluated by
- *means of a set of real numbers, called weights or probabilities, ranging from 0 to 1.*
- To every point in the sample space we assign a probability such that the sum of all probabilities is 1.
- If we have reason to believe that a certain sample point is quite likely to occur when the experiment is conducted, the probability assigned should be close to 1.
- On the other hand, a probability closer to 0 is assigned to a sample point that is not likely to occur.



2.4 Probability of an event

- In many experiments, such as tossing a coin or a die, all the sample points have the same chance of occurring and are assigned equal probabilities.
- For points outside the sample space, that is, for simple events that cannot possibly occur, we assign a probability of 0.
- To find the probability of an event A , we sum all the probabilities assigned to the sample points in A .
- This sum is called the probability of A and is denoted by $P(A)$.



Definition 2.9:

The **probability** of an event A is the sum of the weights of all sample points in A . Therefore,

$$0 \leq P(A) \leq 1, \quad P(\phi) = 0, \quad \text{and} \quad P(S) = 1.$$

Furthermore, if $A_1, A_2, A_3, \dots$ is a sequence of mutually exclusive events, then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots.$$

Example 2.24

A coin is tossed twice. What is the probability that at least 1 head occurs?

Solution:

The sample space for this experiment is $S = \{HH, HT, TH, TT\}$.

If the coin is balanced, each of these outcomes is equally likely to occur.

Therefore, we assign a probability of ω to each sample point.

Then $4\omega = 1$, or $\omega = 1/4$.

If A represents the event of at least 1 head occurring, then

$$A = \{HH, HT, TH\} \text{ and } P(A) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}.$$



Example 2.25:

A die is loaded in such a way that an even number is twice as likely to occur as an odd number. If E is the event that a number less than 4 occurs on a single toss of the die, find $P(E)$.

Solution: The sample space is $S = \{1, 2, 3, 4, 5, 6\}$.

We assign a probability of w to each odd number and a probability of $2w$ to each even number.

Since the sum of the probabilities must be 1, we have $9w = 1$ or $w = 1/9$. Hence, probabilities of $1/9$ and $2/9$ are assigned to each odd and even number, respectively.

Therefore, $E = \{1, 2, 3\}$ and $P(E) = \frac{1}{9} + \frac{2}{9} + \frac{1}{9} = \frac{4}{9}$. ┘



Example 2.26:

In Example 2.25, let A be the event that an even number turns up and let B be the event that a number divisible by 3 occurs. Find $P(A \cup B)$ and $P(A \cap B)$.

Solution: For the events $A = \{2, 4, 6\}$ and $B = \{3, 6\}$, we have

$$A \cup B = \{2, 3, 4, 6\} \text{ and } A \cap B = \{6\}.$$

By assigning a probability of $1/9$ to each odd number and $2/9$ to each even number, we have

$$P(A \cup B) = \frac{2}{9} + \frac{1}{9} + \frac{2}{9} + \frac{2}{9} = \frac{7}{9} \quad \text{and} \quad P(A \cap B) = \frac{2}{9}.$$



Probability of an event

- If the sample space for an experiment contains N elements, all of which are equally likely to occur, we assign a probability equal to $1/N$ to each of the N (points) elements.
- The probability of any event A containing $\underline{n}$ of these N sample (points) elements is then the ratio of the number of elements in A to the number of elements in S .



Rule 2.3: If an experiment can result in any one of N different equally likely outcomes, and if exactly n of these outcomes correspond to event A , then the probability of event A is

$$P(A) = \frac{n}{N}.$$

Example 2.27: A statistics class for engineers consists of 25 industrial, 10 mechanical, 10 electrical, and 8 civil engineering students. If a person is randomly selected by the instructor to answer a question, find the probability that the student chosen is (a) an industrial engineering major and (b) a civil engineering or an electrical engineering major.

Solution: Denote by I , M , E , and C the students majoring in industrial, mechanical, electrical, and civil engineering, respectively. The total number of students in the class is 53, all of whom are equally likely to be selected.

- (a) Since 25 of the 53 students are majoring in industrial engineering, the probability of event I , selecting an industrial engineering major at random, is

$$P(I) = \frac{25}{53}.$$

- (b) Since 18 of the 53 students are civil or electrical engineering majors, it follows that

$$P(C \cup E) = \frac{18}{53}.$$



Example 2.28: In a poker hand consisting of 5 cards, find the probability of holding 2 aces and 3 jacks.

The number of ways of being dealt 2 aces from 4 cards is

$$\binom{4}{2} = \frac{4!}{2! 2!} = 6,$$

and the number of ways of being dealt 3 jacks from 4 cards is

$$\binom{4}{3} = \frac{4!}{3! 1!} = 4.$$

multiplication rule (Rule 2.1), there are $n = (6)(4) = 24$ hands with 2 aces

and 3 jacks. The total number of 5-card poker hands, all of which are equally likely, is

$$N = \binom{52}{5} = \frac{52!}{5! 47!} = 2,598,960.$$

Therefore, the probability of getting 2 aces and 3 jacks in a 5-card poker hand is

$$P(C) = \frac{24}{2,598,960} = 0.9 \times 10^{-5}.$$



Probability of an Event

- If the outcomes of an experiment are not equally likely to occur, the probabilities must be assigned on the basis of prior knowledge or experimental evidence.
- For example, if a coin is not balanced, we could estimate the probabilities of heads and tails by tossing the coin a large number of times and recording the outcomes.
- According to the relative frequency definition of probability, the true probabilities would be the fractions of heads and tails that occur in the long run.
- Another intuitive way of understanding probability is the indifference approach.
- For instance, if you have a die that you believe is balanced, then using this indifference approach, you determine that the probability that each of the six sides will show up after a throw is $1/6$.



Probability of an event

- To find a numerical value that represents adequately the probability of winning at tennis, we must depend on our past performance at the game as well as that of the opponent and, to some extent, our belief in our ability to win.
- Intuition would undoubtedly also play a part in determining the size of the bet that we might be willing to wager.
- The use of intuition, personal beliefs, and other indirect information in arriving at probabilities is referred to as the subjective definition of probability.



Probability of an event

- In most of the applications of probability in this book, the relative frequency interpretation of probability is the operative one.
- Its foundation is the statistical experiment rather than subjectivity, and it is best viewed as the limiting relative frequency.
- As a result, many applications of probability in science and engineering must be based on experiments that can be repeated.



2.5 Additive Rule

- Often it is easiest to calculate the probability of some event from known probabilities of other events.
- This may well be true if the event in question can be represented as the union of two other events or as the complement of some event.
- Several important laws that frequently simplify the computation of probabilities follow.
- The first, called the additive rule, applies to unions of events.

Theorem 2.7

If A and B are two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$



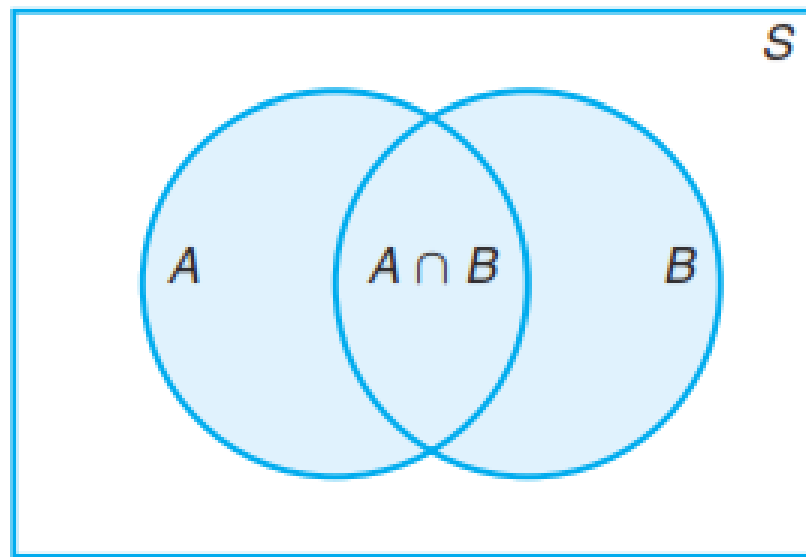


Figure 2.7: Additive rule of probability.

Proof:

Consider the Venn diagram in Figure 2.7. The $P(A \cup B)$ is the sum of the probabilities of the sample points in $A \cup B$.

Now $P(A) + P(B)$ is the sum of all the probabilities in A plus the sum of all the probabilities in B .

Therefore, we have added the probabilities in $(A \cap B)$ twice.

Since these probabilities add up to $P(A \cap B)$, we must subtract this probability once to obtain the sum of the probabilities in $A \cup B$.

Corollary 2.1:

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B).$$

Corollary 2.1 is an immediate result of Theorem 2.7, since if A and B are mutually exclusive, $A \cap B = \emptyset$ and then $P(A \cap B) = P(\emptyset) = 0$.

In general, we can write Corollary 2.2.

Corollary 2.2: If $A_1, A_2, \dots, A_n$ are mutually exclusive, then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n).$$

A collection of events $\{A_1, A_2, \dots, A_n\}$ of a sample space S is called a **partition** of S if $A_1, A_2, \dots, A_n$ are mutually exclusive and $A_1 \cup A_2 \cup \dots \cup A_n = S$. Thus, we have

Corollary 2.3: If $A_1, A_2, \dots, A_n$ is a partition of sample space S , then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n) = P(S) = 1.$$



As one might expect, Theorem 2.7 extends in an analogous fashion.

Theorem 2.8: For three events A , B , and C ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) \\ - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

Example 2.29

John is going to graduate from an industrial engineering department in a university by the end of the semester. After being interviewed at two companies he likes, he assesses that his probability of getting an offer from company A is 0.8, and his probability of getting an offer from company B is 0.6. If he believes that the probability that he will get offers from both companies is 0.5, what is the probability that he will get at least one offer from these two companies?

Solution: Using the additive rule, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.6 - 0.5 = 0.9.$$



Example 2.30

What is the probability of getting a total of 7 or 11 when a pair of fair dice is tossed?

Solution:

Let A be the event that 7 occurs and B the event that 11 comes up.

Now, a total of 7 occurs for 6 of the 36 sample points, and a total of 11 occurs for only 2 of the sample points.

Since all sample points are equally likely, we have $P(A) = 1/6$ and $P(B) = 1/18$.

The events A and B are mutually exclusive, since a total of 7 and 11 cannot both occur on the same toss.

Therefore,
$$P(A \cup B) = P(A) + P(B) = \frac{1}{6} + \frac{1}{18} = \frac{2}{9}.$$

This result could also have been obtained by counting the total number of points for the event $A \cup B$, namely 8, and writing

$$P(A \cup B) = \frac{n}{N} = \frac{8}{36} = \frac{2}{9}.$$



Theorem 2.7

If A and B are two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Corollary 2.1:

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B).$$

Corollary 2.2:

If $A_1, A_2, \dots, A_n$ are mutually exclusive, then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n).$$

Corollary 2.3:

If $A_1, A_2, \dots, A_n$ is a partition of sample space S , then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n) = P(S) = 1.$$

- Theorem 2.7 and its three corollaries should help the reader gain more insight into probability and its interpretation.



Corollary 2.1:

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B).$$

Corollary 2.2:

If $A_1, A_2, \dots, A_n$ are mutually exclusive, then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n).$$

- Corollaries 2.1 and 2.2 suggest the very intuitive result dealing with the probability of occurrence of at least one of a number of events, no two of which can occur simultaneously.
- The probability that at least one occurs is the sum of the probabilities of occurrence of the individual events.



Corollary 2.3:

If $A_1, A_2, \dots, A_n$ is a partition of sample space S , then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n) = P(S) = 1.$$

- The third corollary simply states that the highest value of a probability (unity) is assigned to the entire sample space S .



Example 2.31

If the probabilities are, respectively, 0.09, 0.15, 0.21, and 0.23 that a person purchasing a new automobile will choose the color green, white, red, or blue, what is the probability that a given buyer will purchase a new automobile that comes in one of those colors?

Solution:

Let G , W , R , and B be the events that a buyer selects, respectively, a green, white, red, or blue automobile. Since these four events are mutually exclusive, the probability is

$$\begin{aligned} P(G \cup W \cup R \cup B) &= P(G) + P(W) + P(R) + P(B) \\ &= 0.09 + 0.15 + 0.21 + 0.23 = 0.68. \end{aligned}$$



- Often it is more difficult to calculate the probability that an event occurs than it is to calculate the probability that the event does not occur.
- Should this be the case for some event A , we simply find $P(A)$ first and then, using Theorem 2.7, find $P(A)$ by subtraction.

Theorem 2.9

If A and A' are complementary events, then

$$P(A) + P(A') = 1.$$

Proof:

Since $A \cup A' = S$ and the sets A and A' are disjoint,

$$P(A \cup A') = P(A) + P(A') = P(S) = 1$$



Example 2.32:

If the probabilities that an automobile mechanic will service 3, 4, 5, 6, 7, or 8 or more cars on any given workday are, respectively, 0.12, 0.19, 0.28, 0.24, 0.10, and 0.07, what is the probability that he will service at least 5 cars on his next day at work?

Solution:

Let E be the event that at least 5 cars are serviced.

$$\text{Now, } P(E) = 1 - P(E'),$$

where E' is the event that fewer than 5 cars are serviced.

Since
$$P(E') = 0.12 + 0.19 = 0.31,$$

it follows from Theorem 2.9 that

$$P(E) = 1 - 0.31 = 0.69.$$



Example 2.33: Suppose the manufacturer's specifications for the length of a certain type of computer cable are 2000 ± 10 millimeters. In this industry, it is known that small cable is just as likely to be defective (not meeting specifications) as large cable. That is, the probability of randomly producing a cable with length exceeding 2010 millimeters is equal to the probability of producing a cable with length smaller than 1990 millimeters. The probability that the production procedure meets specifications is known to be 0.99.

- (a) What is the probability that a cable selected randomly is too large?
- (b) What is the probability that a randomly selected cable is larger than 1990 millimeters?

Solution: Let M be the event that a cable meets specifications. Let S and L be the events that the cable is too small and too large, respectively. Then

- (a) $P(M) = 0.99$ and $P(S) = P(L) = (1 - 0.99)/2 = 0.005$.
- (b) Denoting by X the length of a randomly selected cable, we have

$$P(1990 \leq X \leq 2010) = P(M) = 0.99.$$

Since $P(X \geq 2010) = P(L) = 0.005$,

$$P(X \geq 1990) = P(M) + P(L) = 0.995.$$

This also can be solved by using Theorem 2.9:

$$P(X \geq 1990) + P(X < 1990) = 1.$$

Thus, $P(X \geq 1990) = 1 - P(S) = 1 - 0.005 = 0.995$.

